Untitled3

October 15, 2026

[1]:
```
import pandas as pd
```

[3]:
```
dataset = pd.read_csv("50_Startups_dataset.csv")
```

[4]:
```
dataset
```

[4]:

	Unnamed: 0	R&D Spend	Administration	Marketing Spend	State
0	0	165349.30	136897.90	471784.20	New York
1	1	162597.80	151377.69	443898.63	California
2	2	153441.61	101145.65	407934.64	Florida
3	3	144372.51	118671.95	383199.72	New York
4	4	142107.44	91391.87	366168.52	Florida
5	5	131877.00	99814.81	362861.46	New York
6	6	134615.56	147198.97	127716.92	California
7	7	130298.23	145530.16	323876.78	Florida
8	8	120542.62	148719.05	311613.39	New York
9	9	123334.98	108679.27	304981.72	California
10	10	101913.18	110594.21	229161.05	Florida
11	11	100672.06	91790.71	249744.65	California
12	12	93863.85	127320.48	249839.54	Florida
13	13	91992.49	135495.17	252665.03	California
14	14	119943.34	156547.52	256513.02	Florida
15	15	114523.71	122616.94	261776.33	New York
16	16	78013.21	121597.65	264346.16	California
17	17	94657.26	145077.68	282574.41	New York
18	18	91749.26	114175.89	294919.67	Florida
19	19	86419.80	153514.21	0.10	New York
20	20	76253.96	113867.40	298664.57	California
21	21	78389.57	153773.53	299737.39	New York
22	22	73994.66	122782.85	303319.36	Florida
23	23	67532.63	105751.13	304768.83	Florida
24	24	77044.11	99281.44	140574.91	New York
25	25	64664.81	139553.26	137962.72	California
26	26	75328.97	144136.08	134050.17	Florida
27	27	72107.70	127864.65	353183.91	New York
28	28	66051.62	182645.66	118148.30	Florida
29	29	65605.58	153032.16	107138.48	New York
30	30	61994.58	115641.38	91131.34	Florida

31	31	61136.48	152702.02	88218.33	New York
32	32	63408.96	129219.71	46085.35	California
33	33	55494.05	103057.59	214634.91	Florida
34	34	46426.17	157694.02	210797.77	California
35	35	46014.12	85047.54	205517.74	New York
36	36	28663.86	127056.31	201126.92	Florida
37	37	44070.05	51283.24	197029.52	California
38	38	20229.69	65948.03	185265.20	New York
39	39	38558.61	82982.19	174999.40	California
40	40	28754.43	118546.15	172795.77	California
41	41	27893.02	84710.87	164470.81	Florida
42	42	23641.03	96189.73	148001.21	California
43	43	15505.83	127382.40	35534.27	New York
44	44	22177.84	154806.24	28334.82	California
45	45	1000.33	124153.14	1904.03	New York
46	46	1315.56	115816.31	297114.56	Florida
47	47	0.10	135427.02	0.10	California
48	48	542.15	51743.25	0.10	New York
49	49	0.10	116983.90	45173.16	California

	Profit
0	192261.93
1	191792.16
2	191050.49
3	182902.09
4	166188.04
5	156991.22
6	156122.61
7	155752.70
8	152211.87
9	149760.06
10	146122.05
11	144259.50
12	141585.62
13	134307.45
14	132602.75
15	129917.14
16	126993.03
17	125370.47
18	124267.00
19	122776.96
20	118474.13
21	111313.12
22	110352.35
23	108734.09
24	108552.14
25	107404.44

```
26  105733.64
27  105008.41
28  103282.48
29  101004.74
30   99937.69
31   97483.66
32   97427.94
33   96779.02
34   96712.90
35   96479.61
36   90708.29
37   89949.24
38   81229.16
39   81005.86
40   78240.01
41   77798.93
42   71498.59
43   69759.08
44   65200.43
45   64926.18
46   49490.85
47   42559.83
48   35673.51
49   14681.50
```

[5]:
```
dataset = dataset.drop(columns=["Unnamed: 0"])
```

[6]:
```
dataset
```

[6]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.30	136897.90	471784.20	New York	192261.93
1	162597.80	151377.69	443898.63	California	191792.16
2	153441.61	101145.65	407934.64	Florida	191050.49
3	144372.51	118671.95	383199.72	New York	182902.09
4	142107.44	91391.87	366168.52	Florida	166188.04
5	131877.00	99814.81	362861.46	New York	156991.22
6	134615.56	147198.97	127716.92	California	156122.61
7	130298.23	145530.16	323876.78	Florida	155752.70
8	120542.62	148719.05	311613.39	New York	152211.87
9	123334.98	108679.27	304981.72	California	149760.06
10	101913.18	110594.21	229161.05	Florida	146122.05
11	100672.06	91790.71	249744.65	California	144259.50
12	93863.85	127320.48	249839.54	Florida	141585.62
13	91992.49	135495.17	252665.03	California	134307.45
14	119943.34	156547.52	256513.02	Florida	132602.75
15	114523.71	122616.94	261776.33	New York	129917.14
16	78013.21	121597.65	264346.16	California	126993.03

17	94657.26	145077.68	282574.41	New York	125370.47
18	91749.26	114175.89	294919.67	Florida	124267.00
19	86419.80	153514.21	0.10	New York	122776.96
20	76253.96	113867.40	298664.57	California	118474.13
21	78389.57	153773.53	299737.39	New York	111313.12
22	73994.66	122782.85	303319.36	Florida	110352.35
23	67532.63	105751.13	304768.83	Florida	108734.09
24	77044.11	99281.44	140574.91	New York	108552.14
25	64664.81	139553.26	137962.72	California	107404.44
26	75328.97	144136.08	134050.17	Florida	105733.64
27	72107.70	127864.65	353183.91	New York	105008.41
28	66051.62	182645.66	118148.30	Florida	103282.48
29	65605.58	153032.16	107138.48	New York	101004.74
30	61994.58	115641.38	91131.34	Florida	99937.69
31	61136.48	152702.02	88218.33	New York	97483.66
32	63408.96	129219.71	46085.35	California	97427.94
33	55494.05	103057.59	214634.91	Florida	96779.02
34	46426.17	157694.02	210797.77	California	96712.90
35	46014.12	85047.54	205517.74	New York	96479.61
36	28663.86	127056.31	201126.92	Florida	90708.29
37	44070.05	51283.24	197029.52	California	89949.24
38	20229.69	65948.03	185265.20	New York	81229.16
39	38558.61	82982.19	174999.40	California	81005.86
40	28754.43	118546.15	172795.77	California	78240.01
41	27893.02	84710.87	164470.81	Florida	77798.93
42	23641.03	96189.73	148001.21	California	71498.59
43	15505.83	127382.40	35534.27	New York	69759.08
44	22177.84	154806.24	28334.82	California	65200.43
45	1000.33	124153.14	1904.03	New York	64926.18
46	1315.56	115816.31	297114.56	Florida	49490.85
47	0.10	135427.02	0.10	California	42559.83
48	542.15	51743.25	0.10	New York	35673.51
49	0.10	116983.90	45173.16	California	14681.50

[7]:
```
dataset.isnull().sum()
```

[7]:
```
R&D Spend          0
Administration     0
Marketing Spend    0
State              0
Profit             0
dtype: int64
```

[8]:
```
dataset["Profit"].sum()
```

[8]:
```
np.float64(5600636.959999999)
```

[16]:
```
x = dataset[["R&D Spend","Administration","Marketing Spend"]]
y = dataset[["Profit"]]
```

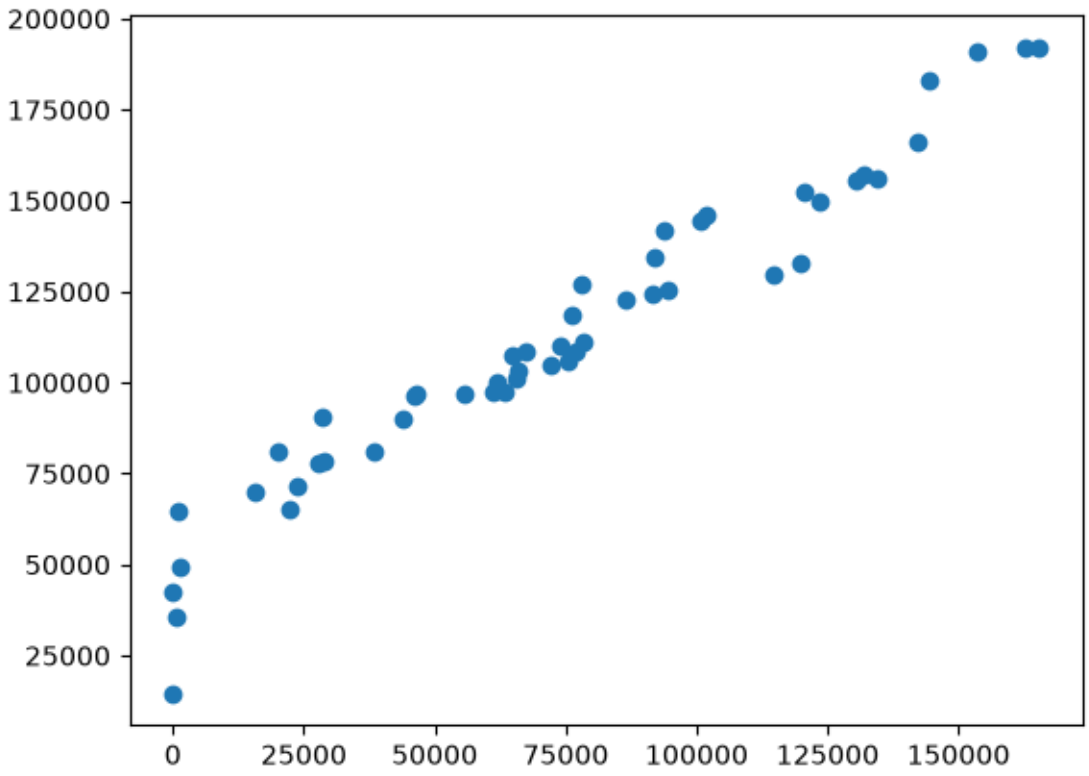
[19]:
```
x
```

[19]:

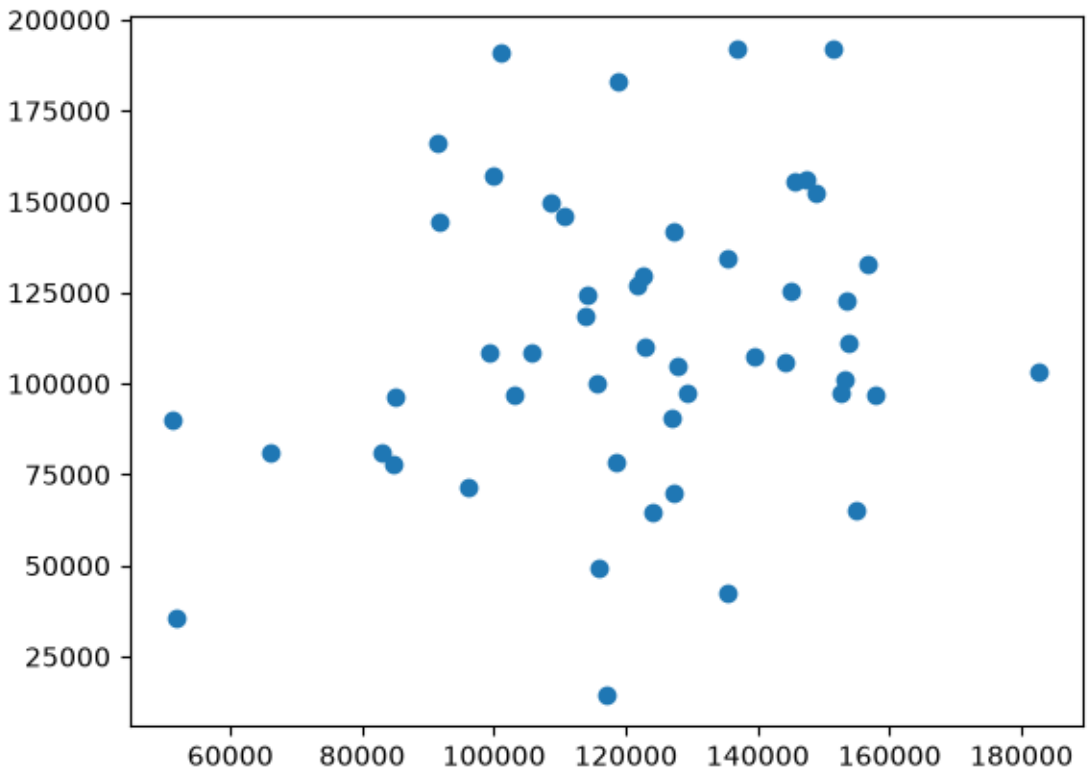
	R&D Spend	Administration	Marketing Spend
0	165349.30	136897.90	471784.20
1	162597.80	151377.69	443898.63
2	153441.61	101145.65	407934.64
3	144372.51	118671.95	383199.72
4	142107.44	91391.87	366168.52
5	131877.00	99814.81	362861.46
6	134615.56	147198.97	127716.92
7	130298.23	145530.16	323876.78
8	120542.62	148719.05	311613.39
9	123334.98	108679.27	304981.72
10	101913.18	110594.21	229161.05
11	100672.06	91790.71	249744.65
12	93863.85	127320.48	249839.54
13	91992.49	135495.17	252665.03
14	119943.34	156547.52	256513.02
15	114523.71	122616.94	261776.33
16	78013.21	121597.65	264346.16
17	94657.26	145077.68	282574.41
18	91749.26	114175.89	294919.67
19	86419.80	153514.21	0.10
20	76253.96	113867.40	298664.57
21	78389.57	153773.53	299737.39
22	73994.66	122782.85	303319.36
23	67532.63	105751.13	304768.83
24	77044.11	99281.44	140574.91
25	64664.81	139553.26	137962.72
26	75328.97	144136.08	134050.17
27	72107.70	127864.65	353183.91
28	66051.62	182645.66	118148.30
29	65605.58	153032.16	107138.48
30	61994.58	115641.38	91131.34
31	61136.48	152702.02	88218.33
32	63408.96	129219.71	46085.35
33	55494.05	103057.59	214634.91
34	46426.17	157694.02	210797.77
35	46014.12	85047.54	205517.74
36	28663.86	127056.31	201126.92
37	44070.05	51283.24	197029.52
38	20229.69	65948.03	185265.20
39	38558.61	82982.19	174999.40
40	28754.43	118546.15	172795.77

41	27893.02	84710.87	164470.81
42	23641.03	96189.73	148001.21
43	15505.83	127382.40	35534.27
44	22177.84	154806.24	28334.82
45	1000.33	124153.14	1904.03
46	1315.56	115816.31	297114.56
47	0.10	135427.02	0.10
48	542.15	51743.25	0.10
49	0.10	116983.90	45173.16

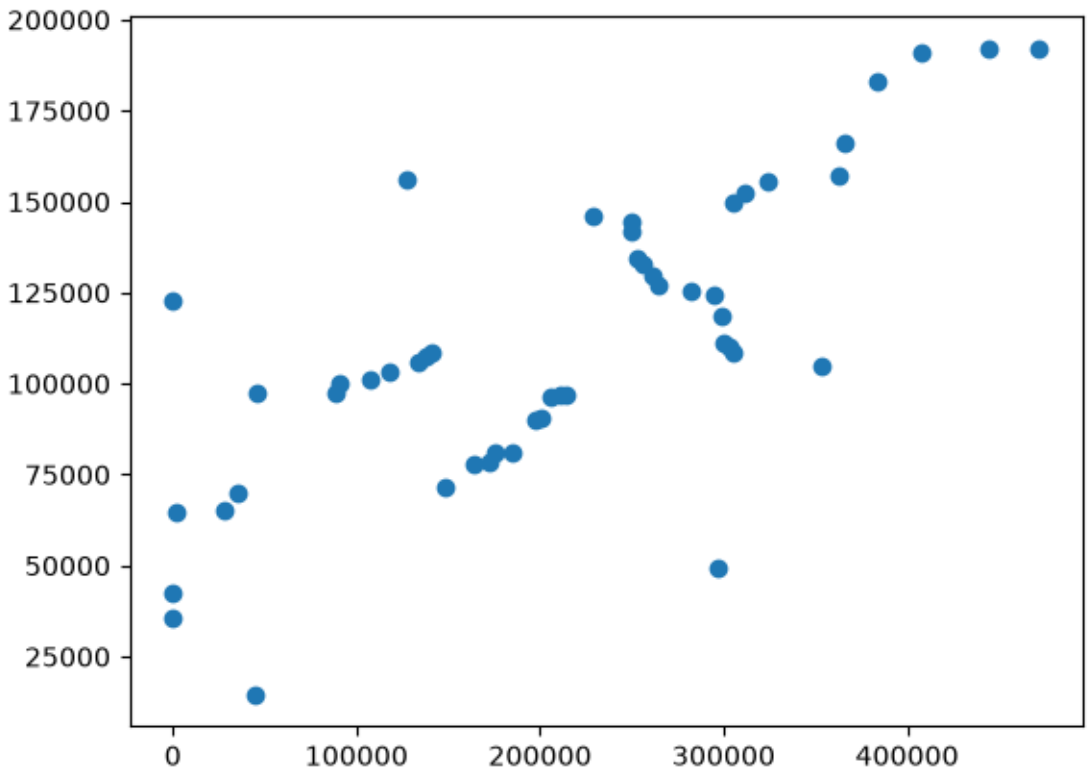
[22]:
```
import matplotlib.pyplot as plt
plt.scatter(x["R&D Spend"],y)
plt.show()
```



[23]:
```
plt.scatter(x["Administration"],y)
plt.show()
```



[24]:
```
plt.scatter(x["Marketing Spend"],y)
plt.show()
```



[25]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size = 0.20,␣
 ↪random_state = 42 )
```

[31]:
```
y_test
```

[31]:

	Profit
13	134307.45
39	81005.86
30	99937.69
45	64926.18
17	125370.47
48	35673.51
26	105733.64
25	107404.44
32	97427.94
19	122776.96

[32]:
```
x_test
```

[32]:

	R&D Spend	Administration	Marketing Spend
13	91992.49	135495.17	252665.03

```
39   38558.61        82982.19        174999.40
30   61994.58       115641.38         91131.34
45    1000.33       124153.14          1904.03
17   94657.26       145077.68        282574.41
48     542.15        51743.25             0.10
26   75328.97       144136.08        134050.17
25   64664.81       139553.26        137962.72
32   63408.96       129219.71         46085.35
19   86419.80       153514.21             0.10
```

[27]:
```
from sklearn.linear_model import LinearRegression
regression= LinearRegression()
regression.fit(x_train,y_train)
```

[27]:
```
LinearRegression()
```

[28]:
```
y_pred= regression.predict(x_test)
y_pred
```

[28]:
```
array([[126703.12716461],
       [ 84894.85081556],
       [ 98893.51815974],
       [ 46501.80815036],
       [129128.49734381],
       [ 50992.79486261],
       [109016.6536578 ],
       [100878.5641454 ],
       [ 97700.69638629],
       [113106.25292226]])
```

[29]:
```
from sklearn.metrics import r2_score
result =r2_score(y_test,y_pred)
result
```

[29]:
```
0.9000653083037319
```

[33]:
```
profit= regression.predict([[25000,30000,32000]])
profit
```

```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\validation.py:2830: UserWarning: X does not have valid
feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

[33]:
```
array([[73128.23517625]])
```

[34]:
```
m = regression.coef_
m
```

[34]:
```
array([[ 0.80377928, -0.06792917,  0.03124155]])
```

[35]:
```
c = regression.intercept_
c
```

[35]:
```
array([54071.89903683])
```

[36]:
```
import pickle
filename ="PROFIT_ML_MODEL.sav"
pickle.dump(regression, open(filename, "wb"))
```

[ ]:
```

```

Untitled4

October 15, 2026

[1]:
```
import pickle
model = pickle.load(open("PROFIT_ML_MODEL.sav", "rb"))
```

[2]:
```
prof = float(input("Enter the R&D spent: "))
prof2 = float(input("Enter the Administration: "))
prof3 = float(input("Enetr the marketing spent: "))
result = model.predict([[prof,prof2,prof3]])
result
```

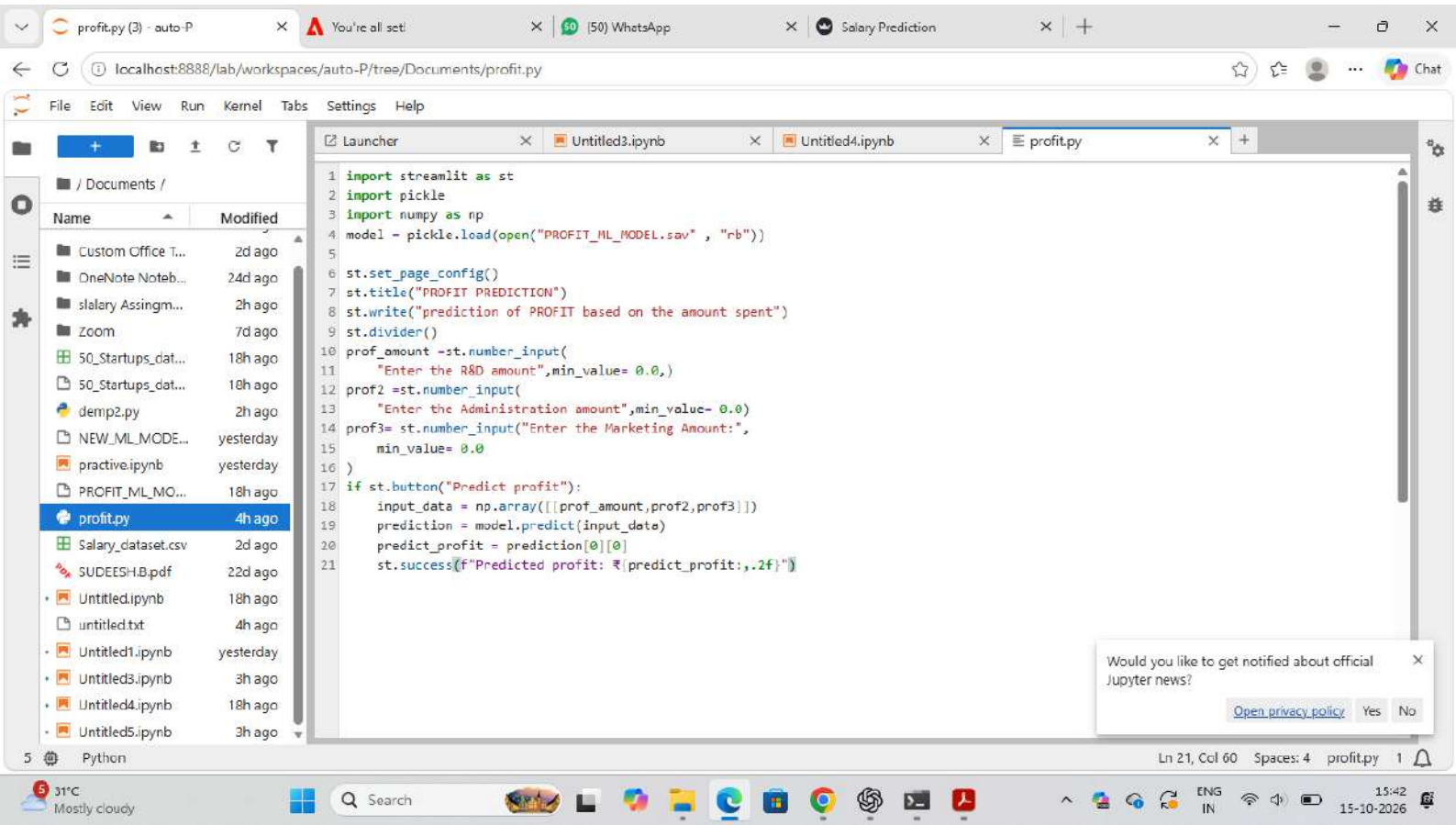
```
Enter the R&D spent:  23000
Enter the Administration:  34000
Enetr the marketing spent:  40000
```

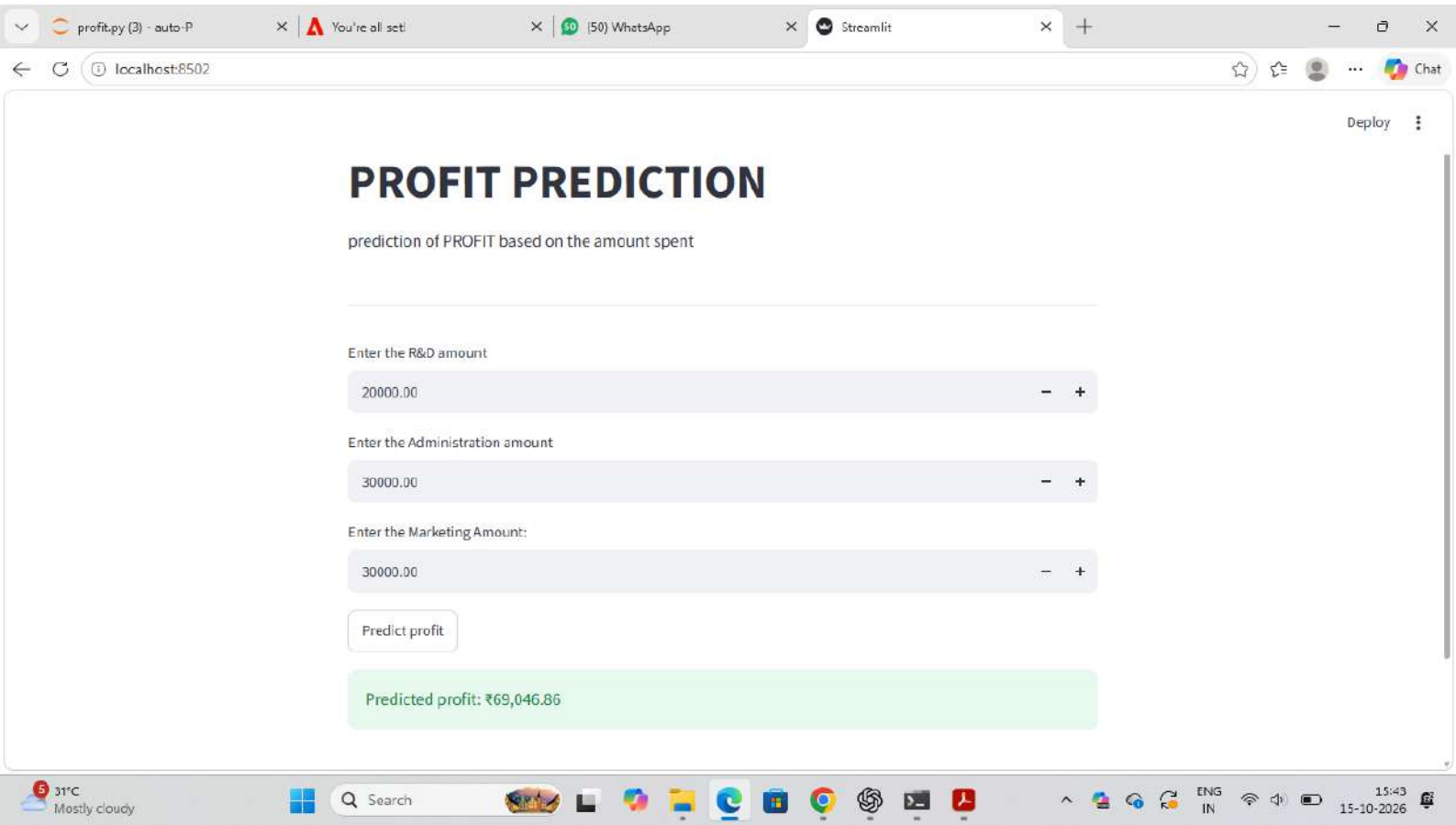
```
C:\Users\acer\AppData\Local\Programs\Python\Python313\Lib\site-
packages\sklearn\utils\validation.py:2830: UserWarning: X does not have valid
feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

[2]:
```
array([[71498.89229617]])
```

[ ]:
```

```


profit.py (3) - auto-P
You're all set!
(50) WhatsApp
Salary Prediction
localhost:8888/lab/workspaces/auto-P/tree/Documents/profit.py
Chat
File Edit View Run Kernel Tabs Settings Help
Launcher
Untitled3.ipynb
Untitled4.ipynb
profit.py
/ Documents /
Name
Modified
Custom Office T... 2d ago
OneNote Noteb... 24d ago
slalary Assingm... 2h ago
Zoom 7d ago
50_Startups_dat... 18h ago
50_Startups_dat... 18h ago
demp2.py 2h ago
NEW_ML_MODE... yesterday
practive.ipynb yesterday
PROFIT_ML_MO... 18h ago
profit.py 4h ago
Salary_dataset.csv 2d ago
SUDEESH.B.pdf 22d ago
Untitled.ipynb 18h ago
untitled.txt 4h ago
Untitled1.ipynb yesterday
Untitled3.ipynb 3h ago
Untitled4.ipynb 18h ago
Untitled5.ipynb 3h ago
import streamlit as st
import pickle
import numpy as np
model = pickle.load(open("PROFIT_ML_MODEL.sav" , "rb"))
st.set_page_config()
st.title("PROFIT PREDICTION")
st.write("prediction of PROFIT based on the amount spent")
st.divider()
prof_amount =st.number_input(
"Enter the R&D amount",min_value= 0.0,)
prof2 =st.number_input(
"Enter the Administration amount",min_value= 0.0)
prof3= st.number_input("Enter the Marketing Amount:",
min_value= 0.0
)
if st.button("Predict profit"):
input_data = np.array([[prof_amount,prof2,prof3]])
prediction = model.predict(input_data)
predict_profit = prediction[0][0]
st.success(f"Predicted profit: ₹{predict_profit:,.2f}")
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python
Ln 21, Col 60 Spaces: 4 profit.py 1
31°C
Mostly cloudy
Search
ENG
IN
15:42
15-10-2026


profit.py (3) - auto-P
You're all set!
(50) WhatsApp
Streamlit
localhost:8502
Chat
Deploy
PROFIT PREDICTION
prediction of PROFIT based on the amount spent
Enter the R&D amount
20000.00
Enter the Administration amount
30000.00
Enter the Marketing Amount:
30000.00
Predict profit
Predicted profit: ₹69,046.86
31°C
Mostly cloudy
Search
ENG
IN
15:43
15-10-2026